The model: a draft

1 Indices and dimensions

	Range	
j	$1, \dots, P$	model parameters
q	$1, \dots, Q$	model species
r	$1, \dots, M$	readouts
s	$1, \dots, S$	scenarios
c	$1, \dots, C$	study cohorts; $s(c)$ is the scenario cohort c was run under
k	$1, \dots, K_{rc}$	statistics that cohort c reports for readout r
i	$1, \dots, N$	simulated patients
$\mathcal{R}_c$	$\subseteq \{1, \dots, M\}$	the readouts that cohort c reports
K_c	$\sum_{r \in \mathcal{R}_c} K_{rc}$	the rows that cohort c contributes
B	$\in \mathcal{B}$	a block: the cohorts that have to be drawn together
K_B	$\sum_{c \in B} K_c$	the rows of block B

$\theta \in \mathbb{R}_+^P$ are the parameters and $\vartheta = \log \theta$. A *target* is a (r, c) pair. Readouts are on the log scale. n_c is the real cohort size, as from the actual data it comes from. A target names a row, not a unit of independence, and neither does a cohort: the block is that. A cohort is the unit that owns an n, an eligibility rule and a level c_{rc}, and most blocks hold exactly one.

2 The model

Population.

$$\vartheta_i \mid \mu, \omega \;\sim\; \mathcal{N}_P(\mu,\, \Omega), \qquad \Omega \;=\; D_\omega\, R\, D_\omega, \qquad D_\omega = \operatorname{diag}(\omega), \tag{1}$$

with R a fixed correlation matrix. Realised with common random numbers, so that ϑ_i is a smooth function of (μ, ω):

$$\vartheta_i \;=\; \mu + D_\omega L_R z_i, \qquad z_i \overset{\text{iid}}{\sim} \mathcal{N}_P(0, I_P) \;\text{ drawn once,} \qquad R = L_R L_R^\top. \tag{2}$$

Mechanism. Simulate patient i under scenario s and read off the species at the measured time:

$$y_i(s) \;=\; g(\vartheta_i;\, s) \;\in\; \mathbb{R}_+^Q. \tag{3}$$

A readout composes those species into the scalar a study reports, on the log scale:

$$x_{irc} \;=\; h_r\big(y_i(1), \dots, y_i(S)\big). \tag{4}$$

Most readouts use only $y_i(s(c))$. A fold change is a contrast between two scenarios in the same patient, which (2) is what makes meaningful, so h_r takes the whole scenario set.

Discrepancy. Two kinds of error, distinguished by where they enter and therefore by how they propagate:

$$\tilde{x}_{irc} \;=\; \kappa_r\Big(h_r\big(\underbrace{e^{\beta}\odot y_i(s(c))}_{\text{mechanism, upstream of } h_r}\big) - c_{rc}\Big) \;+\; c_{rc} \;+\; \gamma_r, \qquad \underbrace{\gamma_r = Z_r^\top a, \quad \log\kappa_r = Z_r^\top b}_{\text{measurement, downstream of } h_r}. \tag{5}$$

γ and κ are the location and scale halves of an affine map on the readout, pivoting about c_{rc}.

$\beta \in \mathbb{R}^Q$ is a log-scale bias on each *species*, shared across readouts and scenarios. It enters upstream of h_r, so it propagates through the composition on its own: a ratio inherits $\beta_{q_1} - \beta_{q_2}$, a percentage inherits $\beta_q - \beta_{\text{nuc}}$, a sum inherits a weighted combination.

a and b are measurement effects on the *readout*, entering downstream of h_r, so they do not propagate. Z_r holds attributes of the readout that have no mechanistic counterpart: assay modality, and whether the quantity is a density, a fraction or a concentration.

$\beta = 0$, $a = 0$, $b = 0$ is a model with no discrepancy, that is, a model that is not biased in scale or location on any of the observed species.

The pivot. c_{rc} is the level readout r sits at in cohort c, taken as that study's own median at (μ_0, ω_0) under its own eligibility rule. It is fixed once and not rebuilt when V_B moves to another ϕ_0: γ_r is the intercept at the reference level, so changing c_{rc} redefines the prior on a.

At $c_{rc} = 0$ the map is $\kappa_r h_r + \gamma_r$, and h_r is a log quantity with a level well away from zero, so κ_r shifts the readout as well as stretching it. Pivoting at c_{rc} leaves κ_r acting on deviations from the study's own level, so it is a spread term within the cohort and γ_r carries the location alone.

The study index adds no parameters: the offset the map applies, $\gamma_r + c_{rc}(1-\kappa_r)$, differs between cohorts only through levels that are already fixed.

The cohort vector of one patient. Stack the corrected readouts that cohort c reports, for a single patient:

$$\tilde{x}_{ic} \;=\; \big(\tilde{x}_{irc}\big)_{r\in\mathcal{R}_c} \;\in\; \mathbb{R}^{|\mathcal{R}_c|}. \tag{6}$$

Built from $\tilde{x}$ and not x: β enters inside h_r, so (5) does not factor through (4).

Cohort selection. Where cohort c reports an eligibility criterion, $w_i^{(c)} \in [0,1]$ is a smooth indicator of whether patient i meets it, and $w_i^{(c)} \equiv 1$ otherwise. A criterion stated as an interval $[\ell_1, \ell_2]$ on readout r gives

$$w_i^{(c)} \;=\; \sigma\big((\tilde{x}_{irc} - \ell_1)/h_w\big)\,\sigma\big((\ell_2 - \tilde{x}_{irc})/h_w\big), \tag{7}$$

with σ the logistic and h_w a fixed bandwidth. Write $\tilde{F}_c$ for the $w^{(c)}$-weighted empirical distribution of $\{\tilde{x}_{ic}\}_{i=1}^N$ on $\mathbb{R}^{|\mathcal{R}_c|}$, and $\tilde{F}_{rc}$ for its r-th marginal.

The criterion is stated in reported units, so it reads $\tilde{x}$ rather than x. Eligibility therefore depends on a, b and β, and is not fixed in ϕ.

Cohorts of one block that share people must share $w^{(c)}$: a common draw serves them all, so criteria that select differently leave it undefined. In practice they are arms and timepoints of one trial and the enrolment criterion is the same.

Statistics. Cohort c reports statistics $T_{rc1}, \ldots, T_{rcK_{rc}}$ for readout r, each a functional of a sample of n_c patients. The model computes the same functionals of samples of the same size drawn from its own predicted population:

$$\tau_{rck}(\phi) \;=\; \mathbb{E}\Big[\, T_{rck}\big(\tilde F^*_{rc}\big) \Big], \qquad \tilde F^*_{rc} = \text{an } n_c\text{-sample from } \tilde F_{rc}. \tag{8}$$

Order the rows of cohort c by $r \in \mathcal{R}_c$ and then by k within r, and stack them into $\tau_c(\phi) \in \mathbb{R}^{K_c}$, and the reported values the same way into $\widehat{T}_c$. Concatenating over $c \in B$ in a fixed cohort order gives τ_B and $\widehat{T}_B$. Since $\tilde F_{rc}$ is a marginal of $\tilde F_c$, the joint form changes V_B and leaves τ alone.

Whatever the source printed is a row, in the form it printed it. Re-expressing a quartile pair as a width would discard the asymmetry between $\hat q_{0.75} - \hat q_{0.5}$ and $\hat q_{0.5} - \hat q_{0.25}$, which is most of what a cohort says about the shape of the pushforward and so about Ω.

The expectation in (8) is what makes τ the mean of (10) rather than the quantity of interest. A statistic computed from n_c patients is not the functional of the population it came from, and the gap closes as n_c grows, not as the predicted cloud does. For quantiles it runs inward on both sides, so predicting the population functional instead would meet the data by shrinking Ω: at a log-scale spread of 1 and $n_c = 9$ the population lower quartile is -0.674 against a reported -0.570, which is recovered at a spread of 0.845. The population functional is the deliverable and is read off the posterior in (22), never off a row.

Smoothing. τ has to be differentiable in ϕ and the reported functionals are not: a quantile picks out one order statistic, so its gradient rests on a single patient and jumps when two patients swap rank. Quantiles are taken with a Beta kernel in cumulative-weight space instead,

$$\hat q(p) \;=\; \frac{\sum_i k_i\, x_{(i)}}{\sum_i k_i}, \qquad k_i = w_{(i)}\, f_{\mathrm{B}}\big(c_i;\, \varkappa,\, n_c - \varkappa + 1\big), \qquad c_i = \frac{\sum_{l \le i} w_{(l)} - \frac{1}{2} w_{(i)}}{\sum_l w_l}, \tag{9}$$

with $\varkappa$ the order statistic the cohort's estimator selects. Sort n_c draws from any distribution and the rank of the $\varkappa$-th in it is $\mathrm{Beta}(\varkappa, n_c - \varkappa + 1)$ whatever its shape, so this is exactly the expectation (8) asks for and differentiability comes with it. Centre $\varkappa/(n_c + 1)$ and width $\sqrt{p(1-p)/n_c}$ are set by n_c rather than chosen.

$\varkappa$ is a convention: $(n_c - 1)p + 1$ in R and numpy, $(n_c + 1)p$ in SPSS, centring at 0.30 and 0.25 when $n_c = 9$. No source records which it used, so declare it per cohort and report the sensitivity.

A width, a standard deviation or an interquartile range, is not an order statistic and (9) does not reach it. It enters as its log, and that log is what makes κ_r an additive shift on the row.

Observation. One block is one block:

$$\widehat{T}_B \;\sim\; \mathcal{N}_{K_B}\big(\tau_B(\phi),\, V_B\big), \qquad \widehat{T}_B \perp \widehat{T}_{B'} \;\text{ for } B \neq B'. \tag{10}$$

Under the ordering of (8), the diagonal blocks of V_B are the per-target covariances and the off-diagonal blocks carry the dependence between readouts, whether of one cohort or of two that were drawn together.

Blocks. Two cohorts sit in one B when a row depends on both. Either one row is built from both, as a contrast across arms is, or they share people, as a subgroup reported beside the whole does. $\mathcal{B}$ is the partition into connected components under the two relations; $|B| = 1$ recovers a cohort.

Which relation put them there decides the draw. Cohorts joined by a shared row are disjoint sets of people and are resampled independently. Cohorts that share people are resampled once from the block's patient set, each cohort's rows evaluated on its own members.

Sharing need not be nesting: 16 evaluable baseline biopsies against day-21 arms of 9 and 10 admit neither a common n nor a containment. The block carries the union and each cohort keeps its n_c. Merging cohorts is the total-overlap case.

Sharing has to be counted to be drawn, as the counts on the Venn regions of the block's cohorts. Where two studies are known to share patients but neither prints which, the draw falls back to independent, V_B understates those off-diagonals and the fit is overconfident there. Record the block as uncounted and bound the effect by refitting without the second study.

Sampling covariance. V_B is the covariance of the reported statistics of block B under resampling of its patients. It has two ingredients. The first is a bootstrap from the predicted population at a plug-in ϕ_0, held fixed thereafter:

$$V_B^{\text{boot}} \;=\; \underset{*}{\text{Cov}}\Big[\, T_B\big(\tilde{F}_B^*\big)\,\Big]_{\phi_0}, \qquad \tilde{F}_B^* = \text{resample of } n_c \text{ patients from } \tilde{F}_c,\; c \in B, \tag{11}$$

where a resampled patient carries its whole vector $\tilde{x}_{ic}$ from (6). Drawing patients rather than readout values is what moves the dependence structure into V_B. Where cohorts of B share people the draw is common: a patient drawn once carries the same $\tilde{x}_i$ into every cohort it belongs to, and that is the only source of across-cohort correlation. It is predicted at ϕ_0, not measured.

The second is E_B, from the emulator $\widehat{g}$ standing in for g. Define it on the statistics and not on the readouts, because the map from a readout error to a statistic error depends on the functional. Draw L held-out clouds $z^{(\ell)}$ at ϕ_0, evaluate each both ways, and take the covariance of the difference:

$$d_B^{(\ell)} \;=\; \tau_B^{\widehat{g}}\big(\phi_0; z^{(\ell)}\big) \;-\; \tau_B^{g}\big(\phi_0; z^{(\ell)}\big), \qquad E_B \;=\; \underset{\ell}{\text{Cov}}\big[\, d_B^{(\ell)}\,\big], \qquad \ell = 1, \ldots, L. \tag{12}$$

This is $K_B \times K_B$ by construction. A covariance is centred, so E_B carries none of the emulator's constant error. Report $\bar{d}_B$ beside it: that constant is a misfit no entry of V_B explains, and (5) is what has to take it up, into a where it can and into μ where it cannot. Section 7 says how much of it (5) can reach.

The fallback is $E_B = \text{diag}(\varepsilon^2)$, with ε_r the held-out residual of readout r repeated across its K_{rc} statistics. It asserts that the emulator errs independently across readouts, which readouts sharing a species do not satisfy, and that a readout error enters a scale statistic with the same loading as a location statistic, where the true loading is near zero.

Where $\widehat{g}$ is trained. E_B and $\bar{d}_B$ are properties of $\widehat{g}$, and $\widehat{g}$ is a property of the parameter values it was trained on. So that training design is an input to the model, not an implementation detail.

The fit calls the emulator at $\vartheta_i = \mu + D_\omega L_R z_i$. The μ part ranges over the stage-1 prior and the z_i part spreads patients around it, so the values it is asked about have covariance about $\Sigma_1 + \Omega$. Draw the design from that law as a scrambled Sobol sequence, widened to cover the flanks and to allow for s making the population wider than ω_0. Every parameter has to vary: a design that pinned one would leave the likelihood undefined along it rather than merely flat.

Score the emulator on an independent draw from the same law, not on a split of the design, since a low-discrepancy sequence leaves its halves interleaved.

Assembling V_B. Since reported uncertainties go to width rather than to the error bar, both the scales and the correlations come from the bootstrap and there is nothing to assemble:

$$V_B \;=\; V_B^{\text{boot}} \;+\; E_B. \tag{13}$$

Sampling noise from the predicted cloud, emulator error on top. A ridge ϵI is added before factoring, since two readouts of one cohort sharing a denominator make V_B near singular.

The cost is that V_B now rests entirely on that cloud. A cloud that is too narrow makes every error bar too small, the likelihood then overrides the prior, and nothing in the fit catches it. Check first, against any source that reports a sampling uncertainty U_{rck} for row (r, c, k) on the log scale:

$$\varrho_{rck} \;=\; U_{rck} \Big/ \sqrt{\left(V_c^{\text{boot}}\right)_{rck,rck}}. \tag{14}$$

This is the residual of a standard-error row taken at ϕ_0 instead of at ϕ, so computing it before fitting is a prior predictive check on the rows that now carry ω. Above one the cloud is too narrow at that readout. Fix the model rather than widening V_B, since widening hides what is being tested.

One number, one job. A reported uncertainty could either set the error bar on its row or say something about population width. It cannot do both, since that counts one number twice. It always goes to width.

The reason is scarcity. Numbers that speak to width are rare, and $\mathcal{M}$ is empty, so the scale rows are all ω has. Sampling noise is abundant by comparison, because (11) supplies it on every row for nothing. So U leaves V_B and becomes a row of $\widehat{T}_B$, and D_B always comes from the bootstrap.

What is predicted depends on which number the source printed, so the extraction still has to record that. A standard deviation is a population spread and is read straight off $\tilde{F}_{rc}$. A standard error is a sampling spread, predicted at ϕ rather than frozen at ϕ_0. At $n_c = 10$ the two differ by a factor of 3.2, so confusing them is not small error.

A predicted standard error needs a smooth estimator. The closed forms serve for most rows: $\text{sd}/\sqrt{n_c}$ for a mean, and $1/(2f(m)\sqrt{n_c})$ for a median, with f the cloud's own density. Both are asymptotic and poor at the smallest n_c, where a bootstrap of (11) at ϕ is better. Cost runs with n_c, so bootstrapping only the small cohorts is close to free. Freeze the resample indices alongside $z_{1:N}$ so the row stays smooth in ϕ.

Such a row buys precision on ω without touching the alias, since κ_r scales a sampling standard deviation exactly as it scales an interquartile range. It is approximate either way, because it depends on how the source built its interval.

Choosing ϕ_0. Whether a plug-in is good enough is checkable: rebuild V_B at other ϕ_0 and read how far the posterior moves. Section 5 supplies one alternative, $\phi_0 = (\hat{\mu}_{\text{flat}},\ \omega_0)$, and it is not automatically the better one. Moving the plug-in moves where $\widehat{g}$ is read, and the emulator's design covers some regions more densely than others, so a plug-in closer to the truth can carry a larger $\bar{d}$. Build V_B at (μ_0, ω_0) by default and use the flat plug-in as the comparison.

3 Priors

Location.

$$\mu \sim \mathcal{N}_P(\mu_0, \Sigma_1), \tag{15}$$

μ_0 and Σ_1 the mean and covariance of the stage-1 composite prior on the log scale. Recall that this is the stage 1- fitting in which we are specifically trying to learn the location parameter per QSP parameter.

Spread. Let $\mathcal{M}$ be the parameters whose between-patient spread a source has measured, with sample size n_j. For $j \in \mathcal{M}$,

$$\log \omega_j \sim \mathcal{N}\left(\log \omega_{0j}, \frac{1}{2(n_j - 1) + \tau^{-2}}\right). \tag{16}$$

For $j \notin \mathcal{M}$,

$$\log \omega_j = \log \omega_{0j} + s + u_j, \qquad s \sim \mathcal{N}(0, \tau_s^2), \qquad u_j \sim \mathcal{N}(0, \tau_\omega^2), \qquad \sum_{j \notin \mathcal{M}} u_j = 0. \tag{17}$$

s is how much wider or narrower the population is overall, and u is the pattern across parameters. Without the constraint those two overlap exactly: add a constant to s, subtract it from every u_j, and ω is unchanged. The constraint removes that flat direction and leaves s as the only global level.

Keeping s clean matters twice over. It is the quantity that reaches the virtual population, and it is the one whose split against b_1 the last simplification below is about.

In practice u is drawn iid and its mean subtracted, which is the same distribution as conditioning on the constraint.

Measurement discrepancy. Free, because it has no parameter twin. Z has an intercept, so a_1 and b_1 are the global location and width corrections.

$$a \sim \mathcal{N}(0, \sigma_a^2 I), \qquad b \sim \mathcal{N}(0, \sigma_b^2 I), \qquad \sigma_a = \sigma_b = 0.5. \tag{18}$$

Mechanism discrepancy. Tightly shrunk, because it competes with μ:

$$\beta_q = \tau_\beta(\beta_q^{\text{raw}} - \bar{\beta}^{\text{raw}}), \qquad \beta_q^{\text{raw}} \sim \mathcal{N}(0,1), \qquad \tau_\beta = 0.15 \text{ fixed}, \tag{19}$$

for q in a declared subset $\mathcal{S} \subset \{1, \ldots, Q\}$, and $\beta_q = 0$ outside it. The default $\mathcal{S}$ is the shared denominators, nucleated_total and V_T.

τ_β is fixed, not estimated. Estimating it creates a funnel, and with β centred there are too few free directions left for the hierarchy to pool anything in return.

β is centred to remove an exact alias. Suppose every species carried the same bias, $\beta_q = c$. Then e^c cancels out of every fraction and every ratio, and shifts only the densities. The measurement map can produce exactly that, by moving γ. So the common level of β is invisible to the data, and the prior alone would decide it. Centring takes it out.

4 Posterior and deliverable

$$\phi \;=\; \big(\mu,\, s,\, u^{\mathrm{raw}},\, \log\omega_{\mathcal{M}},\, a,\, b,\, \beta^{\mathrm{raw}}_{\mathcal{S}}\big). \tag{20}$$

μ is sampled as $\mu = \mu_0 + L_1\mu^{\mathrm{raw}}$ with $\Sigma_1 = L_1 L_1^{\top}$. Where $\mathcal{M}$ is empty the $\log\omega_{\mathcal{M}}$ block is absent, and (16) never applies.

$$p(\phi \mid \mathcal{D}) \;\propto\; p(\phi) \prod_{c=1}^{C} \mathcal{N}\big(\widehat{T}_B;\, \tau_B(\phi),\, V_B\big). \tag{21}$$

The virtual population is the posterior predictive over patients,

$$p(\theta_{\mathrm{new}} \mid \mathcal{D}) \;=\; \int \mathcal{N}_P\big(\log\theta_{\mathrm{new}};\, \mu,\, \Omega\big)\; p(\phi \mid \mathcal{D})\, d\phi, \tag{22}$$

and it is emitted as an ensemble: one population per posterior draw of ϕ, so that between-patient variability and uncertainty about ϕ stay separable in whatever is computed downstream.

5 The flat fit

The flat fit estimates the centre of the population and holds its shape fixed. Set $\omega = \omega_0$ and drop the three spread terms from the parameter vector:

$$\phi_{\mathrm{flat}} \;=\; \big(\mu,\, a,\, \beta^{\mathrm{raw}}_{\mathcal{S}}\big), \qquad \omega = \omega_0, \qquad b = 0. \tag{23}$$

Nothing else changes. Equations (2) to (21) run as written, with ω a constant rather than a function of ϕ.

It is fitted by optimisation, with a Gauss-Newton covariance at the optimum rather than a posterior sample. What this fit has to deliver is a point, the plug-in, plus enough uncertainty in μ to carry into the width shortfall. Both come out of that covariance.

Why ω_0 and not zero. Setting $\omega = 0$ collapses the cloud onto one point, and three things die with it. $\tilde{F}_c$ becomes a point mass, so (11) returns zero and V_B has to come from outside the model. $w^{(c)}$ becomes a switch on the whole cohort instead of on patients, so the eligibility rule stops working and turns discontinuous in μ. And every location functional of a point mass returns the same number, so a cohort that reports a mean beside a median predicts one value for two rows.

Each of those is a diagnostic, and diagnostics are what the flat fit is for. The third is the sharpest: the gap between a reported mean and a reported median is the skewness of the pushforward, which at $\omega = \omega_0$ the model predicts rather than absorbs.

What it buys.

V_B is computable	(11) resamples a real cloud, so the bootstrap works unchanged.
The scale rows are held out	Predict them at ω_0 and report the ratio per readout. That is the width shortfall, before any population parameter is estimated.
$\hat{\mu}$ needs no translation	It is already a population centre, so there is no Jensen gap to correct afterwards.
It supplies the plug-in	Take $\phi_0 = (\hat{\mu}_{\mathrm{flat}},\, \omega_0)$ for (11).

Fit the location rows and hold the scale rows out, so the ratio above is an honest comparison.

b is pinned at zero here, because location rows cannot see it. The pivot sits at the median, so κ_r drops out of a median row entirely, and it enters a mean row only through the small gap between the mean and the median. The rows that carry b are the scale rows, which this fit holds out by design. b is estimated in the population fit.

Cost. A gradient step still costs $C \times N$ forward passes, but N only has to resolve the location statistics and a rough width, not the flanks at a precision that supports inference on ω, so it sits far below the N of the population fit.

What it cannot test. The shape of the population: ω, R, and the direction s and b_1 share. It does test the prior on μ, the forward map, the discrepancy terms, the block structure of (10) and the sampler, and a fault in any of those is one the population fit would also have.

6 What the rows determine

ϕ has far more coordinates than there are rows, so most of the posterior is the stage-1 prior handed back. That is the design rather than a failure. Stage 1 exists to supply what these rows cannot. What is worth knowing is which directions the rows did move, since that is what separates the two.

Linearise τ at the plug-in, whiten by V, and standardise by the prior:

$$G = I + J^\top J, \qquad J = V^{-1/2} \frac{\partial \tau}{\partial \phi} \Sigma_\phi^{1/2}. \tag{24}$$

J has K rows, so $J^\top J$ has rank at most K, and at least $\dim \phi - K$ of its eigenvalues are zero. Along one of those the posterior precision is I, which is the prior. The reason is direct: if $Jv = 0$ then moving ϕ along v leaves every predicted row exactly where it was, so no data at these rows could distinguish the two.

Mixing in the forward map does not change this. It decides which combinations are constrained, not how many.

This is a statement about the eigenbasis and not about single parameters. A given μ_j projects partly onto informed directions and partly onto flat ones, so it usually returns a little narrower than its prior rather than identical. It is also local: the flat directions turn as ϕ moves. Report the spectrum of G and the split of $\mathrm{tr}(J^\top J)$ across the blocks of ϕ.

Two consequences follow from the shape alone.

The measurement map takes most of the information. a has a handful of coefficients against hundreds of mechanism parameters, and it acts directly on the rows where the mechanism has to act through the simulator. So these rows mostly estimate a. Whatever the model can route into a will end up there, the emulator offset of §7 included.

Row space is saturated. When $\mathrm{rank}(J)$ equals the number of rows, any question of the form "can the model produce this row effect" is answered yes by construction. That is why (27) asks what an effect costs instead of whether it is reachable.

The spectrum also says what the sampler faces. A wide spread means trajectories run much further along the loosest direction than the tightest, which warmup will not find on its own. G^{-1} at the mode is cheap and can be given to the sampler as its metric.

7 Checks

Three quantities to compute before the population fit is believed. All are cheap next to the fit itself.

Is the predicted cloud wide enough? The ratio ϱ_{rck} of (14) at ϕ_0, and the same comparison in §5 against scale rows no fit has seen.

Can the measurement map hold the emulator's offset? To first order at $a = b = 0$ the map moves a row by

$$\frac{\partial \tau_{rck}}{\partial a} = Z_r^\top \mathbb{1}\{k \text{ location}\}, \qquad \frac{\partial \tau_{rck}}{\partial b} = Z_r^\top \mathbb{1}\{k \text{ scale}\}, \tag{25}$$

a common shift leaving an interquartile range alone, and (5) pivoting where a location statistic sits. Stack these over all $K = \sum_c K_c$ rows into W and report how much of $\bar{d}$ survives projection onto them, whitened:

$$\text{captured} \;=\; \left\|P_{V^{-1/2}W}\, V^{-1/2}\bar{d}\right\| \Big/ \left\|V^{-1/2}\bar{d}\right\|. \tag{26}$$

High says the emulator's bias is representable, and paid for in a. Low says it is sitting in μ.

Does Z have work of its own? (5) asserts that assay modality and quantity type have no mechanistic twin. Asking whether the mechanism *can* reproduce the row effect of a column of Z is vacuous at $P \gg K$, since $\partial\tau/\partial(\mu, \omega)$ then spans row space and the residual is zero whatever Z is. Ask what it costs instead:

$$\delta_j^* \;=\; \arg\min_{\delta}\; \delta^\top \Sigma^{-1} \delta \qquad \text{subject to} \qquad J_{\mu\omega}\,\delta \;=\; We_j, \tag{27}$$

with We_j the row effect of a unit coefficient on column j, $J_{\mu\omega}$ the Jacobian of τ in the mechanism, and Σ the prior covariance of $(\mu, \log\omega)$. Report $\|\delta_j^*\|$ in prior standard deviations. Large means only an implausible mechanism move imitates that column, so it is identified against the mechanism. Small means the two are confounded and the priors decide the split.

8 Fixed inputs

R	Population correlation matrix. Not identified by these data.
ω_0	Prior centre of the spreads, per parameter.
μ_0, Σ_1	Stage-1 composite prior, log scale.
h_r	Readout composition, including unit conversions. Already in the target definitions.
Z	Readout attributes for the measurement discrepancy: assay modality, quantity type.
c_{rc}	Pivot of (5): cohort c's median of readout r at (μ_0, ω_0). Fixed once, not rebuilt with V_B.
$\mathcal{S}$	Species allowed a mechanism discrepancy. Default: the shared denominators.
$z_{1:N}$	The latent cloud, drawn once.
$w^{(c)}$	Eligibility weights, read from each paper's criterion.
h_q, h_w	Smoothing bandwidths: the quantile kernel (9) and the eligibility weight (7).
$\widehat{g}$	The emulator, and with it the design it was trained on. Fixes E_B and $\bar{d}_B$.
V_B	Assembled by (13) at ϕ_0, then frozen. One per block of (10).
E_B	Emulator error covariance on the statistics of block B, from (12).
τ_s, τ_ω, τ, σ_a, σ_b	Prior widths.
τ_β	Mechanism discrepancy scale, fixed rather than estimated.

9 Known simplifications

- **Independence across blocks.** Equation (10) asserts that distinct blocks are distinct patients. Shared patients are handled inside a block, so what remains is that no two publications overlap, which registry sources do not guarantee.

  The within-block correlation is predicted rather than measured, since a study reporting order statistics at two timepoints does not report their joint. Where per-patient values are recoverable, as digitised paired plots often make them, V_B^{boot} can be taken from the reported patients instead.

  Report the effective row count $K_B/\{1+(K_B-1)\bar{\rho}_B\}$ at mean off-diagonal correlation $\bar{\rho}_B$. Rows can outnumber patients: one source supplies 20 of the 54 targets from about ten patients per arm, and there V_B^{boot} is singular and only E_B makes V_B invertible.

- **No study offset.** A free η_c would be aliased with the measurement discrepancy. It is omitted and γ_r absorbs the study effect, so γ_r mixes assay bias with study bias.

- V **frozen at** ϕ_0**.** Equation (21) is an estimating equation, not a likelihood. Letting V move with ϕ would reintroduce a $-\frac{1}{2}\log\det V$ term that rewards a narrow population regardless of fit.

- s **and** b_1 **share a direction.** When the fitted population comes out narrow there are two explanations, and the data do not separate them: the patients really are alike, or the simulator under-disperses and κ has stretched the readouts to compensate. Only s reaches the virtual population, since b_1 is discarded with the rest of the measurement map, so the split changes what gets shipped.

  Let λ_m be the share of readout m's across-patient variance coming from parameters in $\mathcal{M}$. A scale row loads on s as $1-\lambda_m$ and on b_1 as 1, so a readout helps only if its λ_m differs from the rest. No parameter has a measured width as the model stands, so $\lambda_m = 0$ everywhere and nothing separates them. τ_s and σ_b decide the split. Report the joint posterior of the pair.

  Measurement error is already carried here by κ, which matches an inflated spread without saying where it came from. Giving it its own term would load on the same rows as b_1 and make the tie three-way. A prior on it would break the tie, and an assay coefficient of variation is easier to come by than a measured between-patient width.

- Z **is asserted, not tested.** The claim that assay modality and quantity type have no mechanistic twin, while species-level bias does, is an argument rather than a result. Projecting a column of Z onto $\mathrm{col}\big(\partial\tau/\partial(\mu,\omega)\big)$ does not test it, for the reason §7 gives. (27) is the version that survives the dimensions, and it has not been run.